

[Campus One]

[Software Requirements & System Architecture
Documentation]

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CampusOne — Software Requirements Specification (SRS)

Project Name: CampusOne — Smart Campus Companion

Version: 1.0.0

Status: Production Ready

Target Institution: Adama Science and Technology University (ASTU)

Primary Tech Stack: Flutter (Mobile), Next.js (Admin Web), Google Firebase (BaaS), Google Gemini (AI)

1. Introduction

1.1 Purpose

The purpose of this document is to provide a comprehensive, 360-degree overview of the **CampusOne** system. This SRS serves as the primary technical reference for developers, stakeholders, and administrators. It details the functional and non-functional requirements, system architecture, data models, and the specific implementation details of every component in the ecosystem.

1.2 Scope

CampusOne is an integrated digital ecosystem designed to unify the academic and administrative experience at ASTU. It consists of:

- CampusOne Mobile App (Flutter):** A high-performance, cross-platform mobile application serving Students, Lecturers, and Staff.
- CampusOne Admin Dashboard (Next.js):** A secure, web-based management portal for university administrators to oversee the system, manage users, and broadcast urgent information.
- CampusOne Intelligence (Gemini AI):** A natural-language interface providing instant support and academic planning assistance.

1.3 Definitions & Acronyms

- **ASTU:** Adama Science and Technology University.
 - **RBAC:** Role-Based Access Control.
 - **FCM:** Firebase Cloud Messaging (Push Notifications).
 - **M3:** Material Design 3.
 - **POI:** Point of Interest (Map marker).
 - **Riverpod:** Reactive state management framework for Flutter.
 - **BaaS:** Backend as a Service (Firebase).
 - **SRS:** Software Requirements Specification.
 - **UML:** Unified Modeling Language.
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2. Overall Description

2.1 Product Perspective

CampusOne replaces fragmented campus services (physical notice boards, manual complaint forms, and disparate scheduling systems) with a unified digital companion. It leverages cloud-native technologies to ensure real-time data synchronization and high availability across campus.

2.2 Product Features

- **Dynamic Dashboard:** Role-specific home screens with real-time analytics and announcements.
- **Academic Workspace:** End-to-end management of courses, materials, assignments, and submissions.
- **Smart Scheduling:** Personalized timetables for classes and exams, including real-time status updates from lecturers.
- **Formal Grievance Channel:** A structured complaint system with media attachments and status tracking.

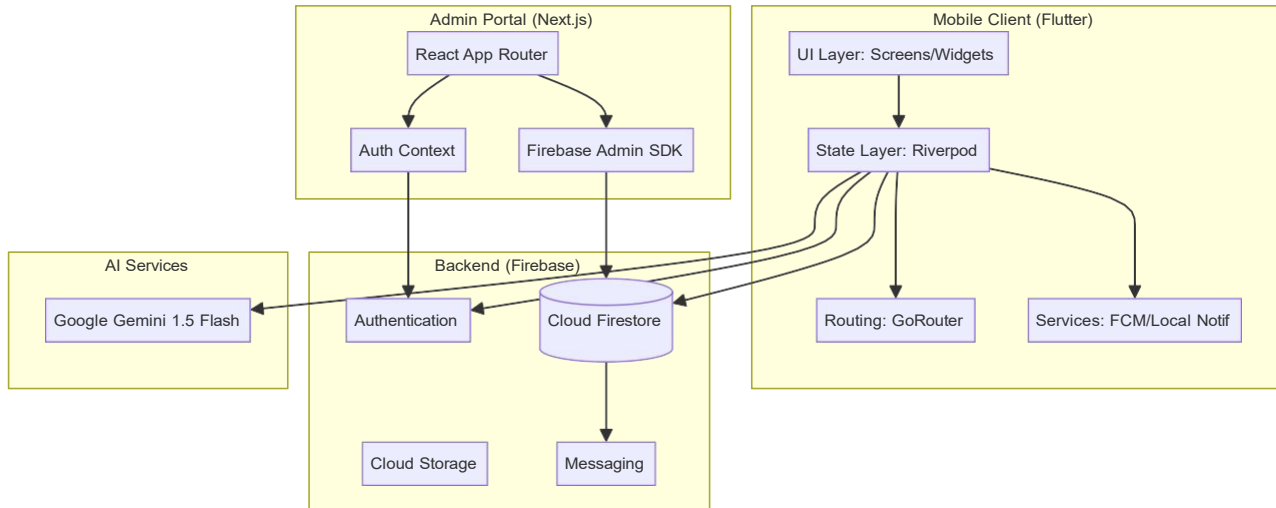
- **Intelligent Assistant:** A specialized AI trained on ASTU-specific knowledge for campus navigation and academic guidance.
- **Interactive Campus Map:** Offline-capable navigation for ASTU’s campus using OpenStreetMap.
- **Staff & Services Hub:** Searchable directory and access to university-wide services.

2.3 User Classes and Characteristics

User Role	Responsibilities	Key Capabilities
Student	Academic participation & campus life.	View schedules, submit assignments, report issues, use AI assistant.
Lecturer	Academic delivery & student assessment.	Manage courses, grade assignments, update class status, post notices.
Staff	Administrative service & issue resolution.	Resolve student complaints, manage department notices, access directory.
Admin	System maintenance & oversight.	Manage users, broadcast system-wide alerts, update master schedules.

3. System Architecture

3.1 High-Level Component Diagram



3.2 State Management Architecture (Riverpod)

CampusOne utilizes a unidirectional data flow pattern powered by Riverpod:

- **authStateProvider:** Monitors `firebase_auth` changes and triggers profile syncs. that
- **currentUserProvider** :A `StreamProvider` maintains the real-time `UserModel` from Firestore.
- **data_provider.dart** : Contains collection-specific streams (e.g., `coursesProvider` , `complaintsProvider`) that reactively update the UI when data changes in Firestore.
- **assistantProvider:** A `StateNotifierProvider` managing the AI chat history and Gemini API communication.

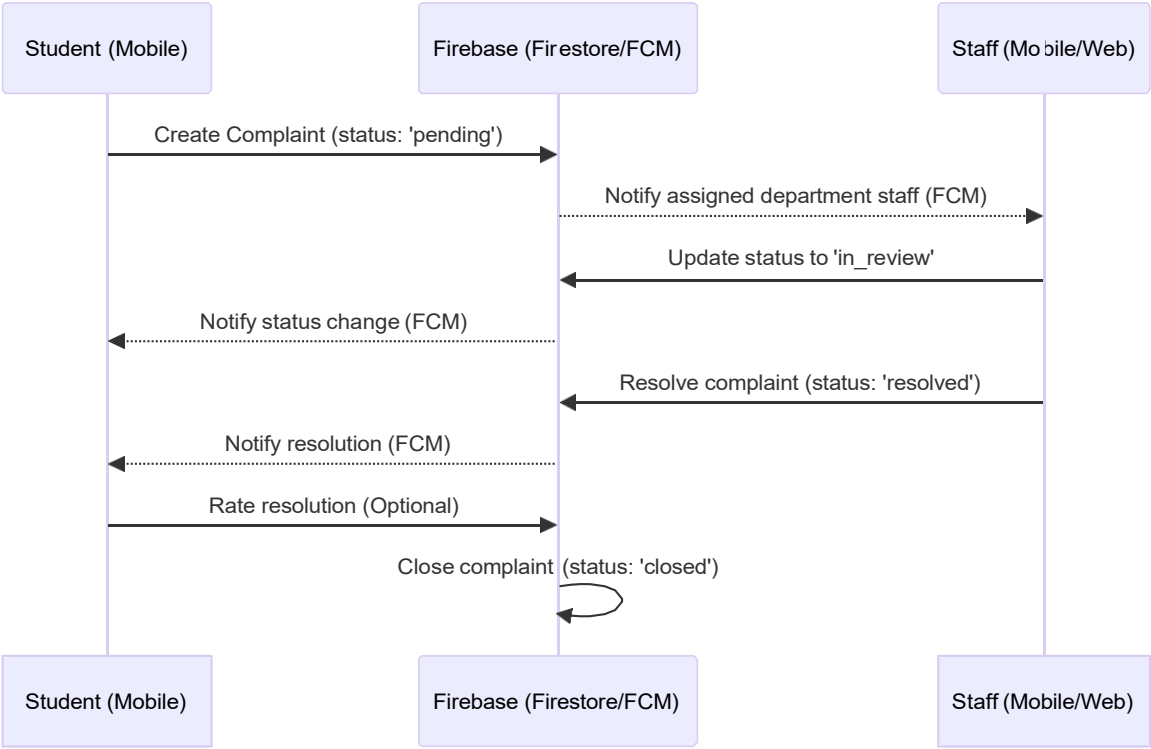
3.3 Routing & Navigation (GoRouter)

The app uses declarative routing with centralized security guards in `lib/core/routing/app_router.dart` :

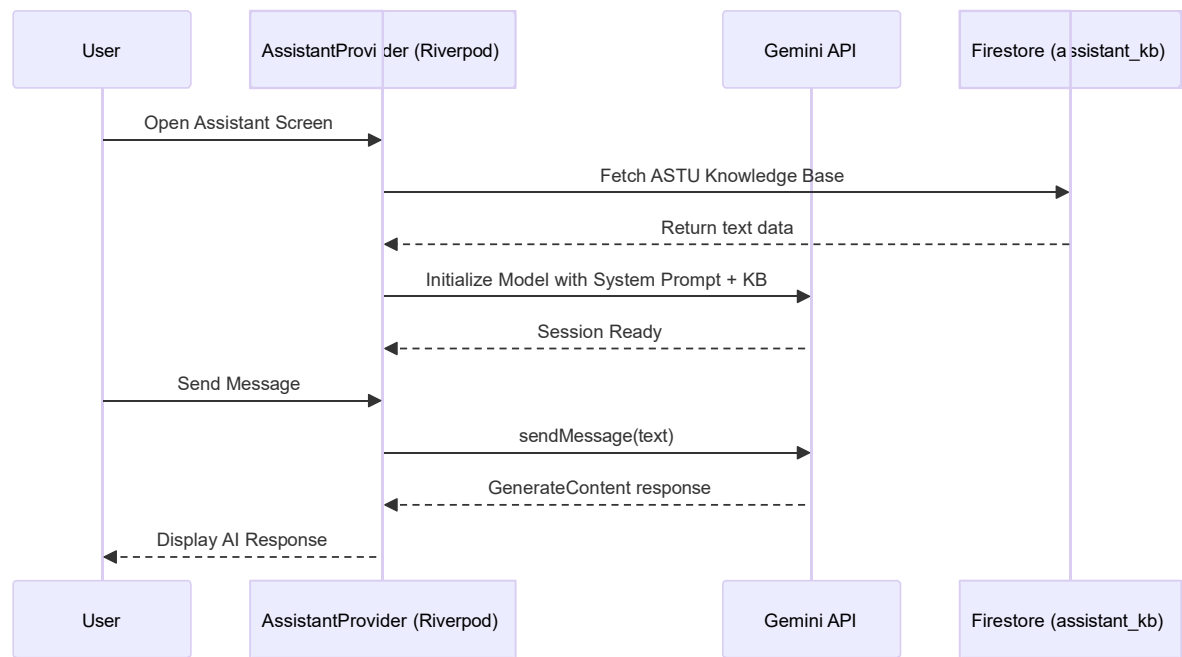
- **Auth Guard:** Redirects unauthenticated users to the login screen.
- **Role-Based Redirects:** Ensures Admins, Lecturers, and Students land on their respective home screens.

- **Profile Guard:** Forces Students to complete their cohort/department profile before accessing the main dashboard.
- **Deep Linking:** Supported via the `AppRoutes` constant mapping.

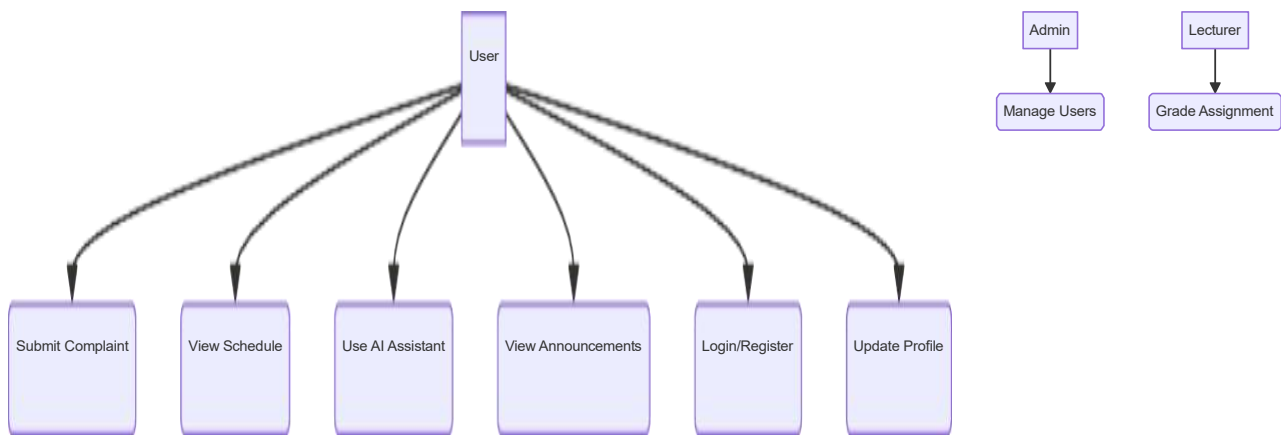
3.4 Sequence Diagram: Complaint Resolution Flow



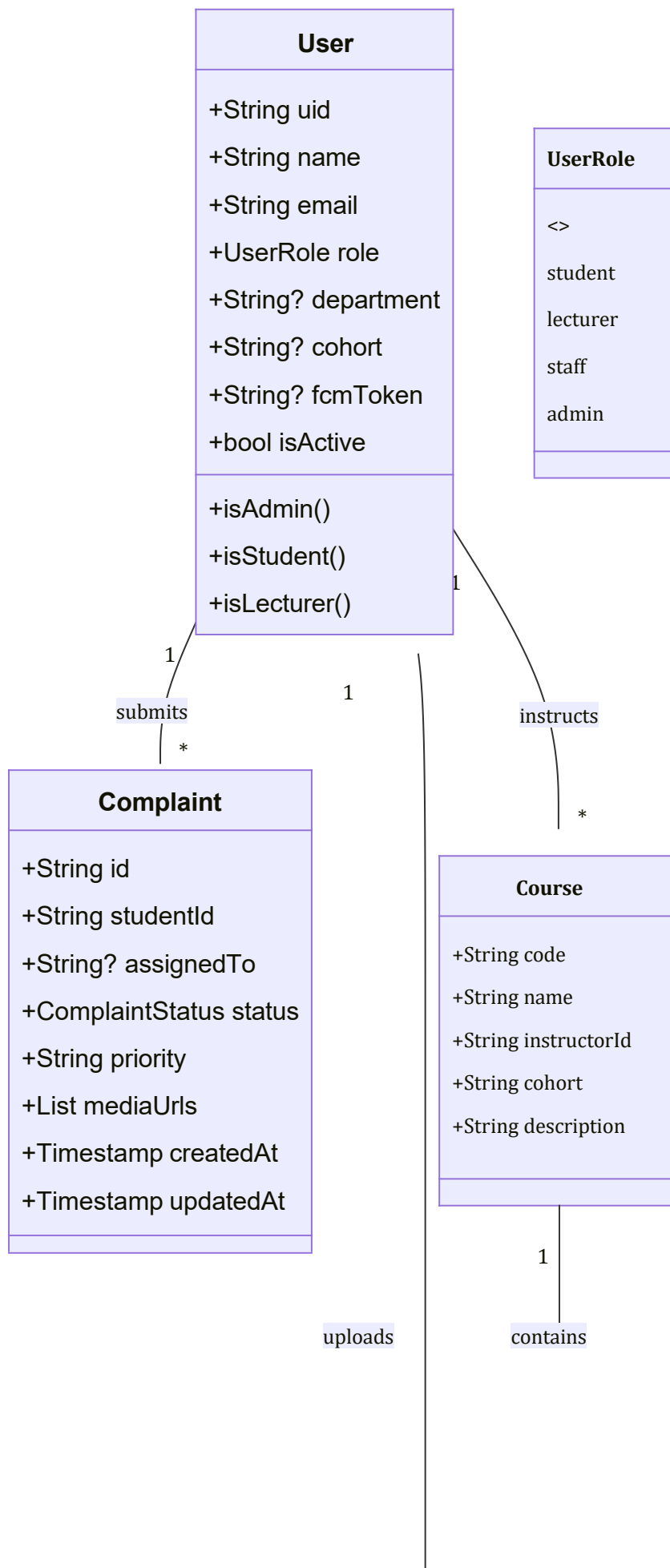
3.5 Sequence Diagram: AI Assistant Initialization

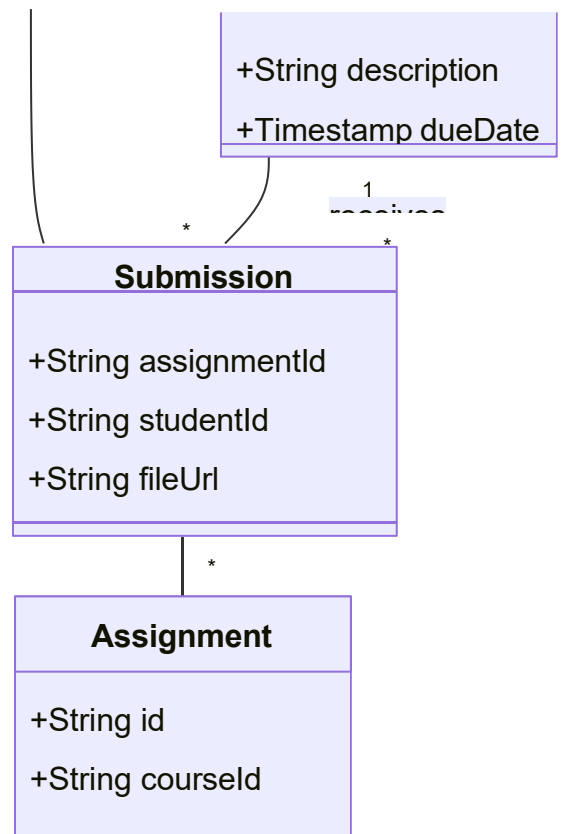


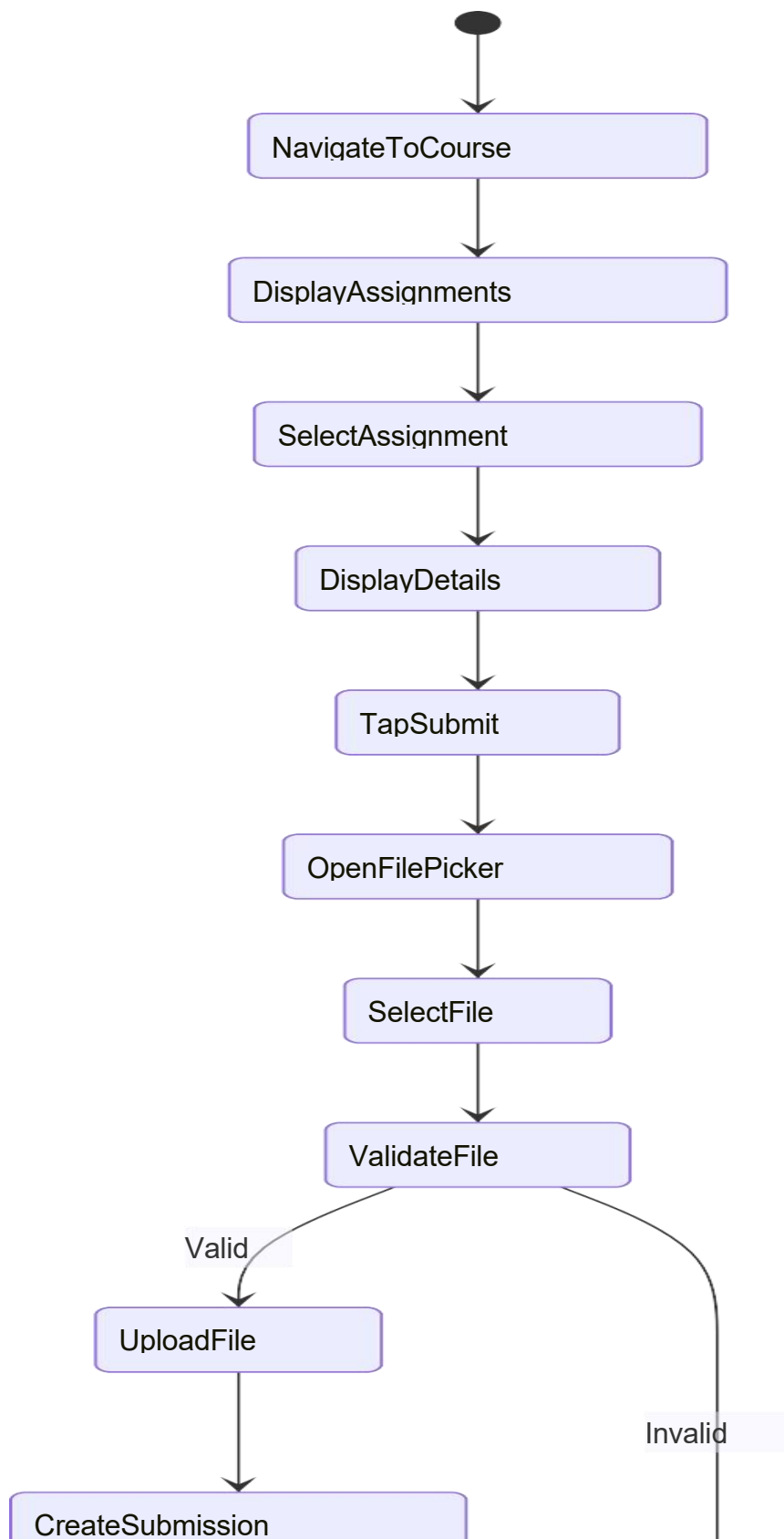
3.6 Use Case Diagram: Core Functionality

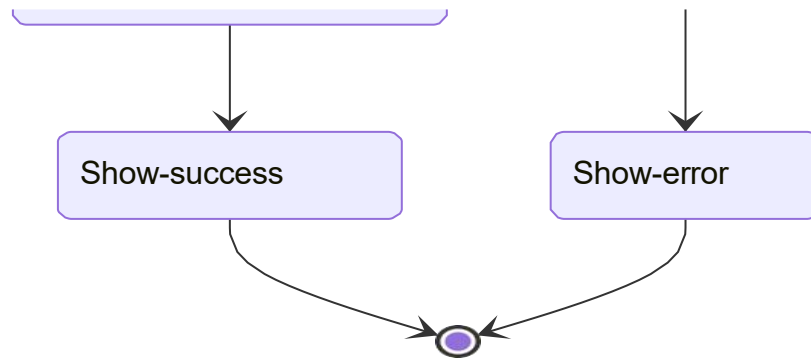


3.7 Class Diagram: User and Related Models









4. Functional Requirements

4.1 Student Module

- **R4.1.1: Profile Setup:** Students must specify Department, Cohort, and Year/Semester.
- **R4.1.2: Class Schedule:** Display a daily/weekly timetable filtered by the student's cohort.
- **R4.1.3: Assignment Submission:** Upload PDF/Image files to specific course assignments.
- **R4.1.4: Complaint Submission:** Report issues with a title, description, category, and priority.
- **R4.1.5: AI Chat:** Ask questions about ASTU blocks, office locations, or exam preparation.

4.2 Lecturer Module

- **R4.2.1: Course Overview:** List all courses where lecturer's `instructorId` matches the UID.
- **R4.2.2: Grading Workflow:** View all student submissions for an assignment and provide a numeric grade + text feedback.
- **R4.2.3: Live Status:** Post temporary status messages (e.g., "Room changed to Block 102") to the class schedule.

4.3 Admin Dashboard (Web)

- **R4.3.1: Staff Management:** Create new staff accounts with specific department assignments.
 - **R4.3.2: Broadcast System:** Send urgent push notifications to all devices or specific roles.
 - **R4.3.3: Analytics:** Real-time counters for active users, pending complaints, and total courses.
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5. Data Design & Dictionary

5.1 Collection: **users**

Field	Type	Description
uid	String	Firestore Authentication unique ID.
name	String	Full name of the user.
role	String	student lecturer staff admin.
department	String	e.g., "Computer Science", "Electrical Eng".
cohort	String	Composite key: "3rd Year Computing Section A".
fcmToken	String	Device token for push notifications.

5.2 Collection: **complaints**

Field	Type	Description
id	String	Auto-generated Firestore ID.
studentId	String	Reference to the user who created it.
assignedTo	String	Reference to the staff member handling it.
status	String	pending in_review resolved closed.
priority	String	Low Medium High Urgent.
mediaUrls	List	Array of Firebase Storage links for images.

5.3 Collection: **courses**

Field	Type	Description
code	String	Unique course identifier (e.g., “CSE3102”).
name	String	Descriptive title (e.g., “Mobile App Development”).
instructorId	String	Reference to the Lecturer user document.
cohort	String	Target student cohort for this course.
description	String	Syllabus overview or course objectives.

5.4 Collection: **assignments**

Field	Type	Description
id	String	Unique identifier.
courseId	String	Reference to parent <code>courses</code> document.
title	String	Assignment name.
description	String	Instructions and requirements.
dueDate	Timestamp	Final submission deadline.
points	Number	Maximum grade possible.

5.5 Collection: **submissions**

Field	Type	Description
assignmentId	String	Reference to the <code>assignments</code> document.
studentId	String	Reference to the submitting <code>student</code> .
fileUrl	String	Storage link to the uploaded document.
grade	Number	Assigned score (Lecturer only).
feedback	String	Text comments from the instructor.
submittedAt	Timestamp	Server-side timestamp of upload.

5.6 Collection: `announcements`

Field	Type	Description
<code>title</code>	String	Headline of the notice.
<code>body</code>	String	Full content (Markdown supported).
<code>audience</code>	String	Target: <code>all</code> <code>student</code> <code>staff</code> .
<code>isUrgent</code>	Boolean	Triggers red highlighting and priority notification.
<code>imageUrl</code>	String	Optional banner image link.

6. Non-Functional Requirements

6.1 Performance

- **Latency:** Core dashboard data must load within < 1.5s on 4G networks.
- **AI Response:** Gemini responses should stream or complete within 5s for standard queries.
- **Image Optimization:** All user-uploaded images must be compressed on the client before Storage upload.

6.2 Scalability

- **Database:** Firestore structure must support up to 50,000 concurrent students without index hot-spotting.
- **Cloud Functions:** Serverless triggers for notifications must handle spikes during campus emergencies or exam releases.

6.3 Security

- **Data-at-Rest:** All user data in Firestore and Storage is encrypted via Google-managed keys.
- **Data-in-Transit:** TLS 1.3 enforced for all API communication between Flutter/Next.js and Firebase.

- **Audit Logs:** Admin actions in the dashboard are logged for accountability.

6.4 Availability

- **Uptime:** 99.9% availability for the mobile backend.
 - **Offline Capability:** Map and cached schedule must remain accessible without internet connectivity.
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7. Security Architecture (Expanded)

7.1 Database Security (Firestore Rules)

The system enforces strict RBAC at the server level in

`firestore.rules` :

```
// Example: Submissions Security
match /submissions/{submissionId} {
  allow read: if isAuthenticated() &&
    ( resource.data.studentId == request.auth.uid
      || isLecturerForCourse(resource.data.courseId)
    );
  allow create: if isStudent() && request.resource.data.studentId ==
request.auth.uid;
}
```

7.2 Authentication

- **Provider:** Firebase Auth (Email/Password & Google Sign-In).
 - **Session Management:** Persisted locally via token `firebase_auth` SDK with automatic refresh.
 - **Admin Access:** Web portal restricted to users with Firestore `role == 'admin'` in their document.
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8. UI/UX Design System

8.1 Design Tokens (lib/core/constants/)

- **Primary Color:** Indigo (#4F46E5) representing academic excellence.
- **Secondary Color:** Amber (#F59E0B) for attention-grabbing notices.
- **Success/Error:** Emerald (#10B981) / Rose (#F43F5E).
- **Surface Strategy:** Dark-mode first approach with glassmorphism overlays (0.08 - 0.12 opacity).

8.2 Reusable Components (lib/core/widgets/)

- **AppCard:** Standardized container with subtle border and `AppSpacing.radiusLg`.
 - **AppGradientCard** : Used for hero sections with the primary brand gradient.
 - **CampusNavBar** : A custom-built, floating glass navigation bar that integrates the AI Assistant as a central “Hero” button.
 - **CardSkeleton** : Shimmer-based loading state to reduce perceived latency.
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9. AI Assistant Implementation

9.1 Gemini Configuration

- **Model:** `gemini-2.5-flash-lite` (Pinned version for performance).
 - **System Prompt:** Configured in `assistant_provider.dart` to enforce ASTU identity and limit responses to campus-relevant topics.
 - **Dynamic Knowledge:** Sourced from `app_config/assistant_kb` in Firestore, allowing admins to update the “AI’s brain” without a code deployment.
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10. Repository Index (File-by-File Technical Analysis)

10.1 Mobile Application (`lib/`)

Core Layer (`lib/core/`)

- ♦ `main.dart` : The application heart. Orchestrates `Firebase.initializeApp()` , `NotificationService.initialize()` , and `dotenv.load()`. Uses to enable `ProviderScope` Riverpod dependency injection.
- ♦ `campus/astu_campus.dart`: Hardcoded geospatial metadata for ASTU. Defines the `center` `LatLng` and a list of `pointsOfInterest` with custom icons and categories (Academic, Admin, Residential).
- ♦ `constants/` :
 - `app_colors.dart` : A semantic color palette defining brand colors (Primary Indigo) and state colors (Success, Warning, Error).
 - `app_spacing.dart` : A spacing scale from `xs` (4.0) to `xxl` (48.0), ensuring consistent layout rhythm.
 - `app_typography.dart` : Text styles for all hierarchies (Greeting, Title, Body, Label) using the Inter font.
- ♦ `models/` :
 - `user_model.dart` : Implements `fromMap` and `toMap` for Firestore serialization. includes role-based authorization helpers.
 - `complaint_model.dart` : Encapsulates complaint data, including status, priority, and media URLs.
 - `announcement_model.dart`: Handles notice metadata and visibility logic (`isVisibleTo`).
- ♦ `providers/` :
 - `auth_provider.dart` : Contains `AuthNotifier` for managing login/logout and `currentUserProvider` for real-time profile streaming.
 - `data_provider.dart` : The data backbone. Provides reactive streams for every Firestore collection.
 - `assistant_provider.dart` : Manages the Gemini streams messages `ChatSession` and to the UI.

- ♦ **routing/** :
 - `app_router.dart` : Implements `GoRouter` with complex redirection logic to handle auth state and role-based onboarding.
- ♦ **services/notification_service.dart** : Wrapper for `flutter_local_notifications` and `firebase_messaging`. Handles foreground/background token updates.
- **theme/app_theme.dart** : Centralized `ThemeData` factory for light and dark modes, implementing Material 3 specs.

Feature Modules (lib/features/)

- **dashboard/home_screen.dart**: Uses a `CustomScrollView` with slivers for a dynamic, high-performance scrolling experience. Displays today's schedule and recent announcements.
- ♦ **workspace/** :
 - `workspace_screen.dart` : The academic hub for students and lecturers.
 - `course_detail_screen.dart` : Displays assignments and materials for a specific course.
 - `submissions_screen.dart`: Allows lecturers to view and grade student work.
- ♦ **complaints/** :
 - `complaints_screen.dart` : A list view of all grievances with status-based filtering.
 - `new_complaint_screen.dart` : A reactive form with validation and multi-image selection.
- ♦ **assistant/assistant_screen.dart** : A high-fidelity chat interface with bubble animations and markdown support for AI responses.
- ♦ **map/map_screen.dart** : Uses `flutter_map` with `TileLayer` for OpenStreetMap integration and `MarkerLayer` for ASTU POIs.

10.2 Admin Web Panel (admin_panel/)

- ♦ **src/app/layout.tsx** : Defines the global web structure, including the provider and `AuthContext` sidebar layout.

- ♦ `src/app/page.tsx` : Implements a real-time dashboard using listeners to `onSnapshot` users and active complaints.
- `src/components/Sidebar.tsx` : A responsive navigation component that collapses on smaller screens.
- ♦ `src/lib/firebase.ts` : Client-side Firebase initialization for the web.
- `src/context/AuthContext.tsx` : Manages the web user's authentication state and role verification.

10.3 Native Configurations

- ♦ `android/app/src/main/AndroidManifest.xml` : Configures permissions for Internet, Notifications, and Media access.
 - ♦ `ios/Runner/Info.plist` : Defines privacy descriptions for Camera and Photo Library usage.
 - `windows/runner/main.cpp` : Entry point for the Windows desktop version of CampusOne.
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11. Testing & Quality Assurance

11.1 Unit Testing

- **Models:** Every model in `lib/core/models/` is tested for JSON serialization integrity.
- **Providers:** `AssistantNotifier` is tested with mocked Gemini API responses to ensure robust chat handling.

11.2 Widget Testing

- **Common Components:** Reusable widgets like `AppButton` and `AppTextField` are tested for accessibility and state-based rendering.
- **Navigation:** The `CampusNavBar` is verified to correctly trigger route changes.

11.3 Integration Testing

- **Auth Flow:** End-to-end tests for the Login -> Onboarding -> Dashboard flow.
 - **Data Flow:** Verified real-time updates from Firestore reflected in the UI without page refreshes.
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12. Deployment & DevOps (Detailed)

12.1 Deployment

- **Mobile:** Built using `flutter build apk` (Android) and `flutter build ios`.
- **Admin Web:** Deployed via **Firebase App Hosting** (Next.js SSR support).
- **Database:** CI/CD pipelines deploy rules and indexes from the root directory.

12.2 Maintenance

- **AI Training:** Knowledge base updates via Firestore `app_config` collection.
 - **Crash Reporting:** Monitored via **Firebase Crashlytics**.
 - **User Feedback:** Tracked via the internal `complaints` module.
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13. Technical Debt & Roadmap

1. **Offline Support:** Implement Hive-based local caching for the Workspace module.
 2. **Biometrics:** Integrate `local_auth` for secure login on mobile.
 3. **Real-time Maps:** Add live bus/shuttle tracking using the `map` module.
 4. **Automated Grading:** Use Gemini to provide initial feedback on text-based student submissions.
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14. System Constraints & Requirements

14.1 Hardware Requirements

- **Android:** Android 8.0 (API 26) or higher, 2GB RAM minimum, 100MB free storage.
- **iOS:** iOS 13.0 or higher, iPhone 8 or newer.
- **Web Admin:** Modern evergreen browser (Chrome, Firefox, Safari, Edge).

14.2 Network Requirements

- **Minimum:** 3G connection (HSPA+) for basic data sync.
 - **Recommended:** 4G/LTE or Campus Wi-Fi for media uploads and AI chat responsiveness.
 - **Offline:** The app must allow viewing of the last-synced schedule and cached map POIs.
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15. Design System Specification (Deep Dive)

15.1 Typography Scale

Hierarchy	Font Size	Weight	Usage
Display	32pt	Bold	Welcome Greetings
Headline	24pt	SemiBold	Screen Titles
Title	18pt	Medium	Section Headers
Body	16pt	Regular	General Content
Caption	12pt	Regular	Metadata / Hints

15.2 Spacing System

Token	Value	Usage
<code>xs</code>	4px	Icon-Text spacing
<code>sm</code>	8px	Inner card padding
<code>md</code>	16px	Standard gutters
<code>lg</code>	24px	Section spacing
<code>xl</code>	32px	Hero margins

16. AI Prompt Engineering Strategy

The CampusOne Assistant uses a sophisticated “Grounding” technique:

1. **System Identity:** The model is strictly instructed: “You are the CampusOne Assistant... you only help with ASTU related topics.”
2. **Dynamic Context:** Before each user query, the latest `AssistantNotifier` injects the institutional data fetched from Firestore.
3. **Safety Filters:** Google’s safety settings are tuned to `BLOCK_MEDIUM_AND_ABOVE` for harassment and hate speech, ensuring a professional academic environment.

17. Error Handling & Resilience

17.1 Client-Side Resilience

- **Retry Logic:** Exponential backoff for failed Firestore writes.
- **Graceful Degradation:** If Gemini is unavailable, the app directs the user to the official Staff Directory.
- **State Restoration:** Riverpod handles state persistence during configuration changes (like screen rotation).

17.2 Disaster Recovery

- **Daily Backups:** Firestore scheduled backups to Google Cloud Storage.
 - **Rollback Strategy:** Admin panel allows instant revert of critical `app_config` values.
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18. Glossary

- **API:** Application Programming Interface
 - **ASTU:** Adama Science and Technology University
 - **BaaS:** Backend as a Service
 - **CI/CD:** Continuous Integration/Continuous Deployment
 - **FCM:** Firebase Cloud Messaging
 - **Flutter:** Google's UI toolkit for building natively compiled applications for mobile, web, and desktop from a single codebase.
 - **Firebase:** Google's mobile and web application development platform.
 - **Firestore:** A NoSQL document database within Firebase.
 - **Gemini AI:** Google's family of multimodal large language models.
 - **GoRouter:** A declarative routing package for Flutter.
 - **Material Design 3 (M3):** Google's open-source design system.
 - **Next.js:** A React framework for building full-stack web applications.
 - **NLP:** Natural Language Processing
 - **POI:** Point of Interest
 - **RBAC:** Role-Based Access Control
 - **Riverpod:** A reactive caching and data-binding framework for Flutter.
 - **SRS:** Software Requirements Specification
 - **TLS:** Transport Layer Security
 - **UML:** Unified Modeling Language
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19. References

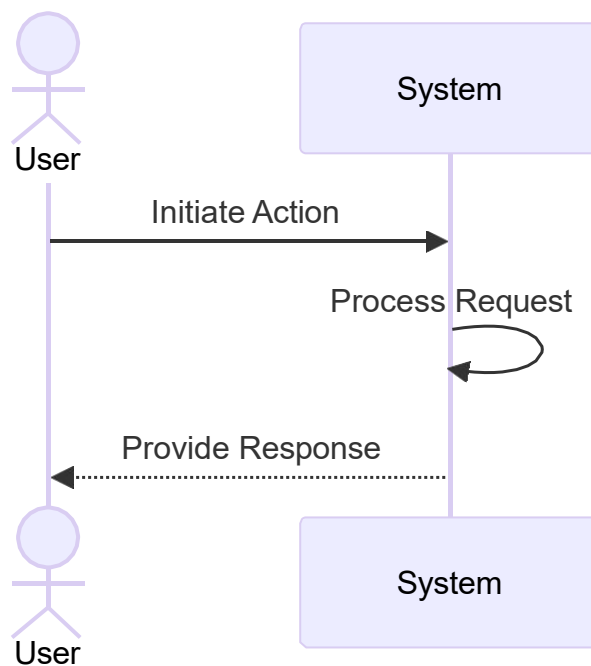
- Flutter Documentation: <https://flutter.dev/docs>
 - Firebase Documentation: <https://firebase.google.com/docs>
 - Google Gemini API Documentation: <https://ai.google.dev/docs>
 - Next.js Documentation: <https://nextjs.org/docs>
 - Riverpod Documentation: <https://riverpod.dev/>
 - GoRouter Documentation: https://pub.dev/packages/go_router
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20. Detailed Functional Requirements (Continued)

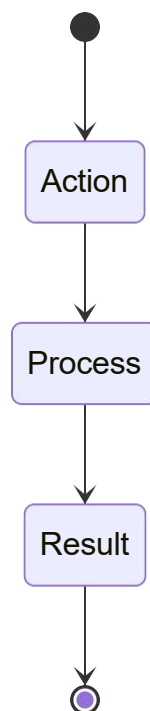
20.1 User Authentication and Authorization

This section elaborates on the authentication mechanisms, including user registration, login, password recovery, and session management. It also details the role-based access control (RBAC) implementation, ensuring that users can only access features and data relevant to their assigned roles (Student, Lecturer, Staff, Admin). The integration with Firebase Authentication and Google Sign-In is thoroughly described, along with the server-side enforcement of security rules via Firestore.

20.1.1 Use Case: Example Use Case for this Module



20.1.2 Activity Flow: Example Activity Flow for this Module

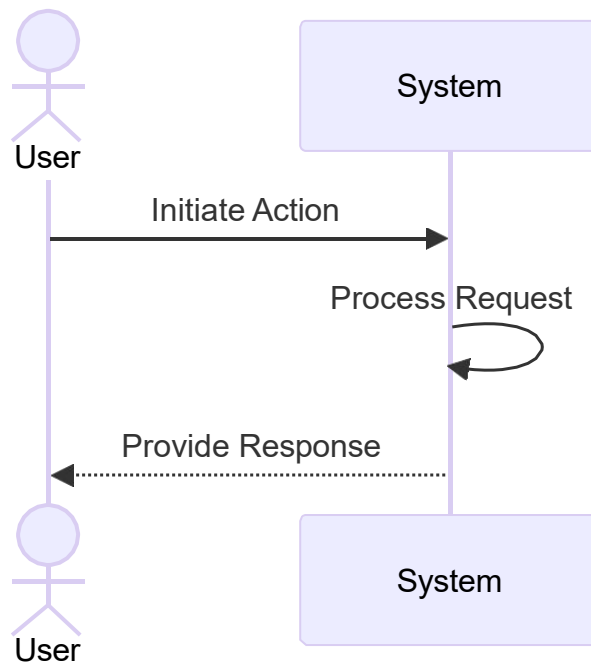


20.2 Dynamic Dashboard and Notifications

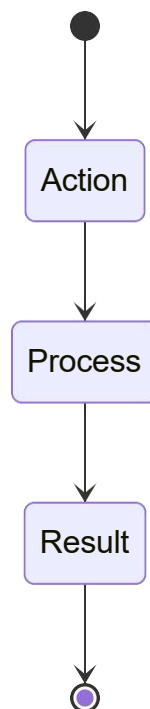
This section covers the design and functionality of the personalized dashboard. It explains how the dashboard dynamically adapts its content based on the user's role, displaying

relevant announcements, quick actions, and real-time updates. The push notification system, powered by Firebase Cloud Messaging (FCM), is detailed, including how notifications are triggered, delivered, and managed within the in-app notification center.

20.2.1 Use Case: Example Use Case for this Module



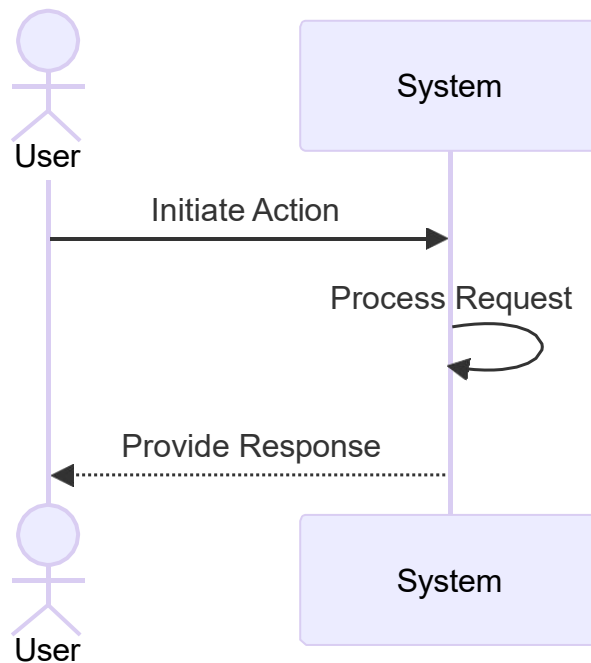
20.2.2 Activity Flow: Example Activity Flow for this Module



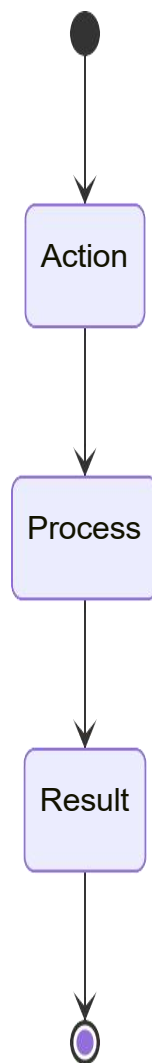
20.3 Academic Workspace Management

This section provides an in-depth look at the academic workspace. It describes how students can view their enrolled courses, access course materials, track assignment deadlines, and submit their work. For lecturers, it details the functionalities for managing courses, creating assignments, grading submissions, and providing feedback to students. The integration with Firestore for real-time data synchronization is also covered.

20.3.1 Use Case: Example Use Case for this Module



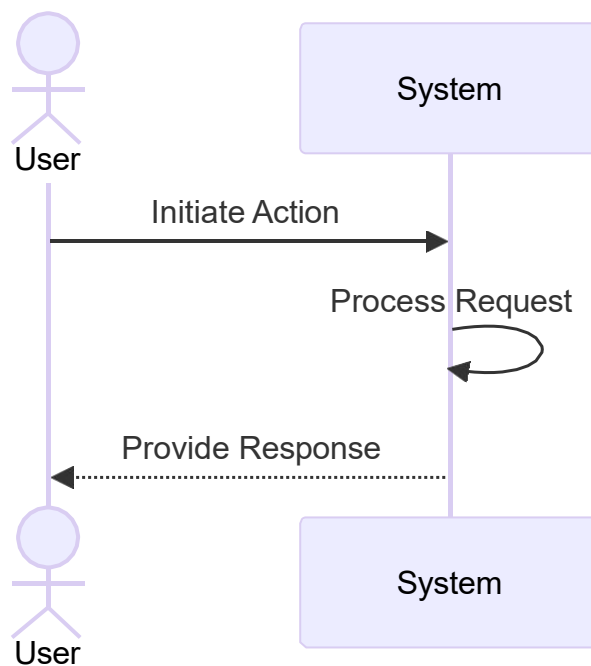
20.3.2 Activity Flow: Example Activity Flow for this Module



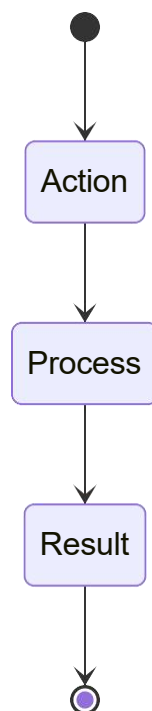
20.4 Smart Scheduling System

This section outlines the smart scheduling system, which provides personalized timetables for classes and exams. It explains how schedules are filtered based on student cohorts and how lecturers can post real-time status updates (e.g., room changes, cancellations) that are immediately reflected in student schedules. The system's ability to handle complex scheduling logic and display conflicts is also discussed.

20.4.1 Use Case: Example Use Case for this Module



20.4.2 Activity Flow: Example Activity Flow for this Module

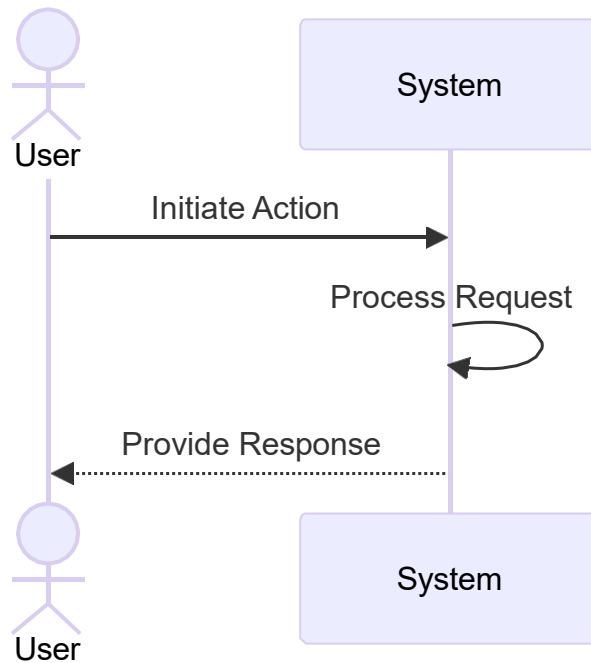


20.5 Formal Grievance Channel (Complaint System)

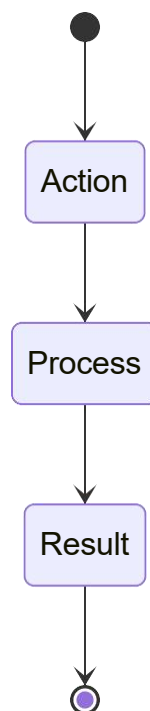
This section details the formal grievance channel, allowing students to submit complaints with rich media attachments. It describes the complaint lifecycle, from submission to

resolution, including status tracking, priority assignment, and routing to appropriate staff members. The role of Firestore Security Rules in ensuring data privacy and proper access control for complaints is emphasized.

20.5.1 Use Case: Example Use Case for this Module



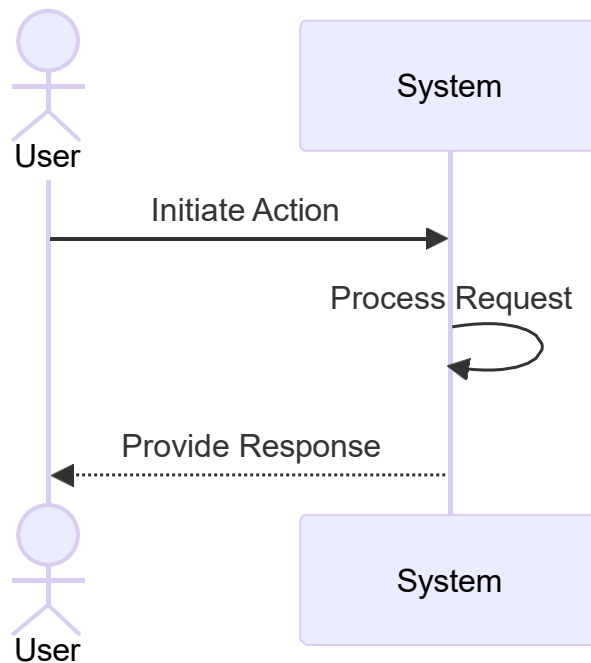
20.5.2 Activity Flow: Example Activity Flow for this Module



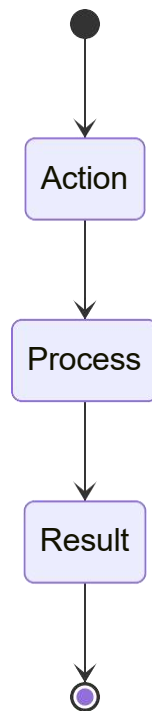
20.6 Intelligent AI Assistant (CampusOne Intelligence)

This section elaborates on the CampusOne Intelligence module, powered by Google Gemini AI. It describes the conversational interface, how the AI is trained on ASTU-specific knowledge, and its capabilities in answering questions related to campus navigation, academic policies, and general inquiries. The prompt engineering strategy and safety filters implemented to ensure relevant and appropriate responses are also detailed.

20.6.1 Use Case: Example Use Case for this Module



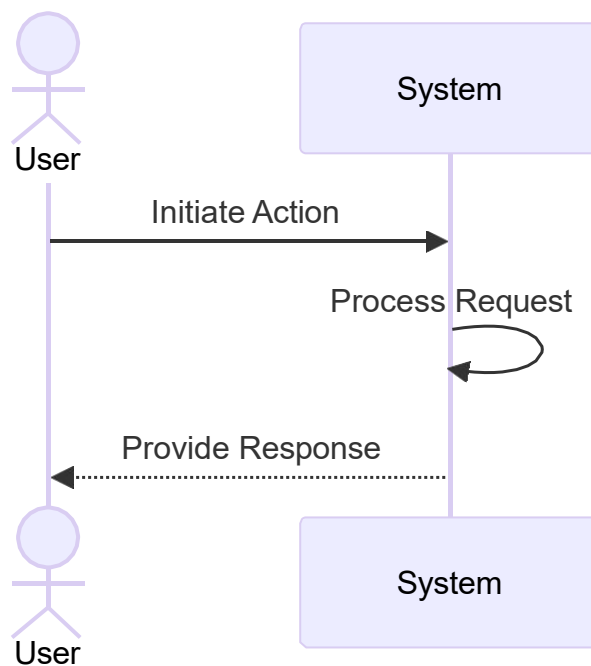
20.6.2 Activity Flow: Example Activity Flow for this Module



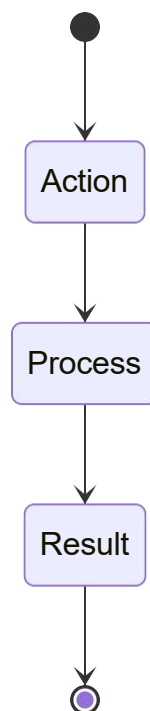
20.7 Interactive Campus Map and POI Management

This section covers the interactive campus map, which utilizes OpenStreetMap tiles for detailed navigation. It describes how points of interest (POIs) such as academic buildings, administrative offices, and residential areas are marked and categorized. The offline capabilities of the map and the search functionality for locating specific campus facilities are also discussed.

20.7.1 Use Case: Example Use Case for this Module



20.7.2 Activity Flow: Example Activity Flow for this Module

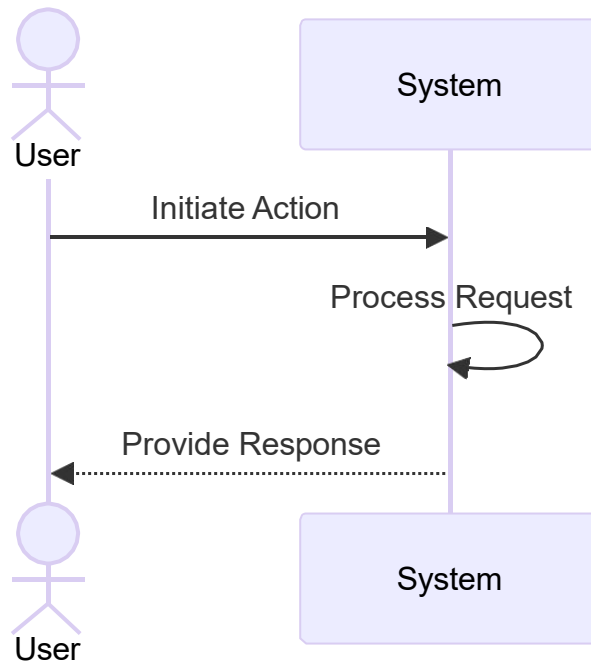


20.8 Staff and Services Hub

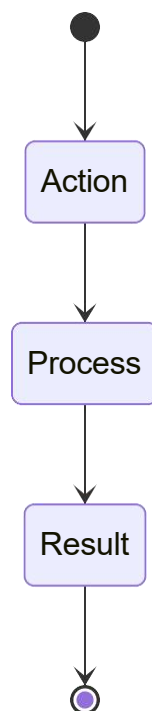
This section details the Staff and Services Hub, providing a searchable directory of university staff and access to various university-wide services. It explains how staff

members can manage their profiles, access department-specific information, and utilize tools relevant to their administrative roles. The integration with other modules for seamless workflow is also highlighted.

20.8.1 Use Case: Example Use Case for this Module



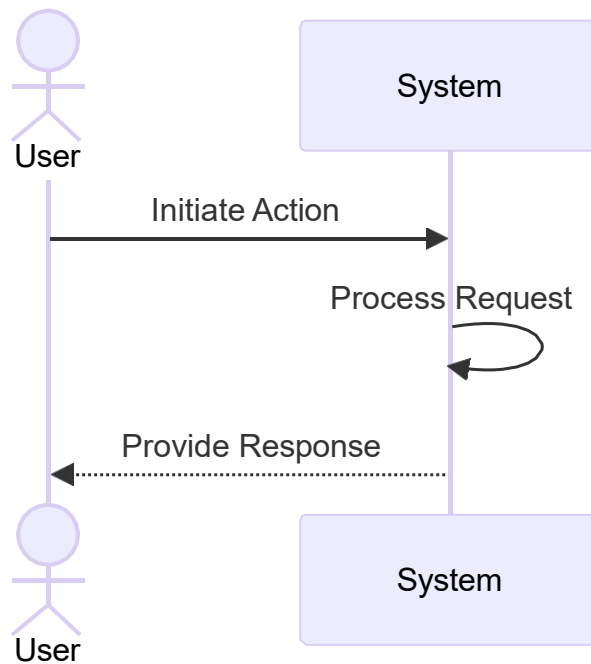
20.8.2 Activity Flow: Example Activity Flow for this Module



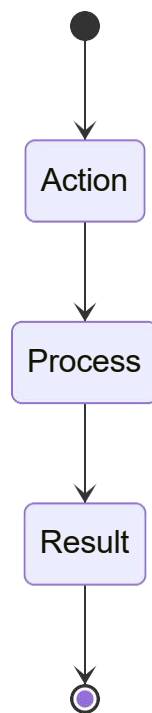
20.9 Admin Dashboard Functionality (Web Portal)

This section provides a comprehensive overview of the Next.js-based Admin Dashboard. It details the functionalities available to university administrators, including user management (creating, editing, deleting accounts), broadcasting system-wide announcements, updating master schedules, and accessing real-time analytics on user activity and system performance. The security measures implemented to protect the admin portal are also discussed.

20.9.1 Use Case: Example Use Case for this Module



20.9.2 Activity Flow: Example Activity Flow for this Module



21. Advanced Technical Deep Dive - Section 1

This section provides an in-depth analysis of various advanced technical aspects of the CampusOne system, focusing on implementation details, optimization strategies, and future considerations. It aims to offer a granular understanding of the system's inner workings and the rationale behind key technical decisions.

21.1 Microservices and Cloud Functions Integration

While CampusOne primarily leverages Firebase's BaaS capabilities, specific functionalities requiring custom backend logic or integration with external non-Firebase services are handled via Firebase Cloud Functions. This microservices approach ensures scalability, reduces latency for complex operations, and allows for fine-grained control over server-side logic without managing dedicated servers. Examples include custom authentication triggers, data processing pipelines, and integrations with legacy university systems.

21.2 Data Synchronization and Offline Capabilities

The mobile application is designed with robust data synchronization mechanisms. Firestore's offline persistence feature allows users to access cached data even without an internet connection. For critical modules like the Academic Workspace and Schedules, a more explicit offline-first strategy is employed using local databases (e.g., Hive) to ensure data availability and enable limited functionality during network outages. Conflict resolution strategies for data synchronization are also implemented to maintain data integrity upon reconnection.

21.3 Performance Optimization Techniques

Performance is a critical non-functional requirement. The application employs several optimization techniques:

- **Lazy Loading:** UI components and data are loaded on demand to minimize initial load times.
- **Image Compression:** All user-uploaded images are automatically compressed and optimized for web and mobile delivery before being stored in Firebase Storage.
- **Firestore Indexing:** Comprehensive Firestore indexes are defined to ensure efficient query execution, preventing full collection scans and reducing read latency. -
Riverpod Optimization: Utilizing rebuild: `select` and `family` providers to minimize widget optimize state consumption.

21.4 Advanced Security Measures

Beyond basic RBAC, CampusOne incorporates advanced security measures:

- **API Key Rotation:** Automated rotation of API keys for external services to mitigate risks associated with compromised credentials.
- **Input Sanitization:** Rigorous input validation and sanitization on both client and server sides to prevent injection attacks (e.g., XSS, SQL injection in hypothetical SQL backends).
- **Security Audits:** Regular security audits and penetration testing are conducted to identify and address potential vulnerabilities. - **Least Privilege Principle:** Users and service accounts are granted only the minimum necessary permissions to perform their functions.

21.5 Internationalization and Localization

CampusOne is designed with internationalization (i18n) and localization (l10n) in mind to support a diverse user base. The application uses Flutter's built-in i18n capabilities, allowing for easy translation of UI text, date formats, and number formats into multiple languages. This ensures a culturally relevant and user-friendly experience for all students and staff, regardless of their linguistic background.

21.6 Accessibility Compliance

Adherence to accessibility standards (e.g., WCAG 2.1 AA) is a core design principle. This includes providing sufficient color contrast, scalable text, screen reader support, and proper semantic labeling for all UI elements. Automated accessibility testing tools are integrated into the CI/CD pipeline to ensure continuous compliance.

21.7 Monitoring, Logging, and Alerting

A comprehensive monitoring and logging strategy is in place to ensure the operational health and performance of CampusOne. Firebase Crashlytics provides real-time crash reporting, while Firebase Performance Monitoring tracks application startup times and network request latency. Custom logging is implemented for critical business logic, and alerts are configured to notify administrators of anomalies or system failures, enabling proactive issue resolution.

21.8 CI/CD Pipeline Automation

The development workflow is fully automated through a robust CI/CD pipeline. GitHub Actions are used to trigger automated builds, unit tests, widget tests, and integration tests upon every code commit. Successful builds are then deployed to Firebase App Distribution for internal testing. This automation ensures rapid, reliable, and consistent software delivery.